



UNIVERSITY OF SRI JAYEWARDENEPURA

Faculty of Technology

Department of Information and Communication Technology

ITS 4202 - Emerging Technologies

Assignment 01 - Designing and Modeling a Smart Agriculture Solution

1. You are tasked with designing a smart agriculture solution for a hypothetical farm facing specific challenges and developing a basic conceptual model to support your design. This project will test your ability to research, analyze, integrate advanced technologies, and create a simple representation of your solution.

Objectives

- Conduct independent research on smart agriculture technologies.
- Develop a tailored solution using IoT, precision agriculture, and advanced technologies.
- Create a basic conceptual model to illustrate your solution.
- Assess potential challenges and propose effective strategies.
- Document your findings in a detailed report.

Task Details

- ✓ • Choose a hypothetical farm scenario (e.g., a region prone to drought, a livestock-focused farm, or an urban vertical farm).

- ✓ • Identify 2-3 key challenges (e.g., water scarcity, disease management, or yield improvement).
- ✓ • Research relevant technologies (e.g., soil sensors, drones, AI analytics) and their real-world applications. Use at least 5 credible sources.
- ✓ • Design a smart agriculture system to tackle the identified challenges.
- ✓ • Specify components (e.g., sensors, actuators, software platforms) and explain their selection.
- ✓ • Create an implementation plan, addressing connectivity, cost, and user adoption.
- ✗ • Develop a basic conceptual model (e.g., a flowchart, diagram, or simple data flow representation) to demonstrate how your system works. This should include key inputs, processes, and outputs.
- ✓ • Evaluate potential obstacles (e.g., data security, return on investment) and suggest mitigation strategies.
- Write a report covering your research, design, model, and analysis.

Deadline: 27th of October 2025 (11.59pm)

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